UNCONVENTIONAL MACHINING PROCESSES

Ultrasonic Machining • Chemical Machining • Electric Discharge Machining • Laser Beam Machining

Elements of Workshop Technology, Vol. II - Hajra Choudhury

1. Introduction to Unconventional Machining

Conventional machining removes material mainly by direct mechanical contact between a harder cutting tool and the workpiece. In many modern engineering situations this becomes difficult or uneconomical: the work material may be extremely hard, brittle, heat resistant, electrically conductive but difficult to cut, or the required feature may be very small, deep or intricate.

Unconventional or non-traditional machining processes use forms of energy other than the ordinary shearing action of a sharp cutting tool. Material may be removed by mechanical erosion with abrasives, controlled chemical dissolution, electrical spark erosion, or intense thermal energy from a laser beam.

These processes are especially valuable for tool steels, carbides, ceramics, glass, superalloys, miniature holes, dies, moulds, slots and complex profiles.

Process	Primary energy / action	Tool-work contact?	Main material requirement
USM	Mechanical vibration + abrasive erosion	No direct cutting contact	Best for hard, brittle materials; electrical conductivity not required
CHM	Controlled chemical dissolution	No	Material must be chemically etchable by selected reagent
EDM	Thermo-electric spark erosion	No physical contact	Workpiece must be electrically conductive
LBM	Focused optical/thermal energy	No	Very broad range; absorption and thermal response matter

2. Ultrasonic Machining (USM)

2.1 Introduction

Ultrasonic Machining is a mechanical-type non-traditional machining process in which a shaped tool is made to vibrate at ultrasonic frequency while an abrasive slurry is maintained in the small gap between the tool and the workpiece. The vibrating tool repeatedly accelerates abrasive particles, which strike the work surface and remove extremely small chips. USM is therefore particularly suitable for hard and brittle materials such as glass, ceramics, carbides, ferrites, quartz and certain hardened materials.

2.2 Principle

The principle of USM is abrasive impact erosion. The tool itself normally does not cut the workpiece by a conventional cutting edge. Instead, the tool oscillates with a very small amplitude at a very high frequency, usually normal to the work surface. Abrasive grains suspended in a liquid carrier are trapped between the tool and workpiece. Repeated hammering of these grains produces high local stresses and causes micro-cracking, chipping and erosion.

- Ultrasonic vibration: frequency above the audible range; industrial systems commonly operate in the tens of kilohertz.
- Small vibration amplitude: typically only a few to a few tens of micrometres.
- Abrasive grains: common abrasives include boron carbide, silicon carbide, aluminium oxide and diamond, depending on application.
- Liquid carrier: water is common; it carries abrasive to the gap and removes debris.
- Shape reproduction: the machined cavity is approximately the negative reproduction of the tool end.

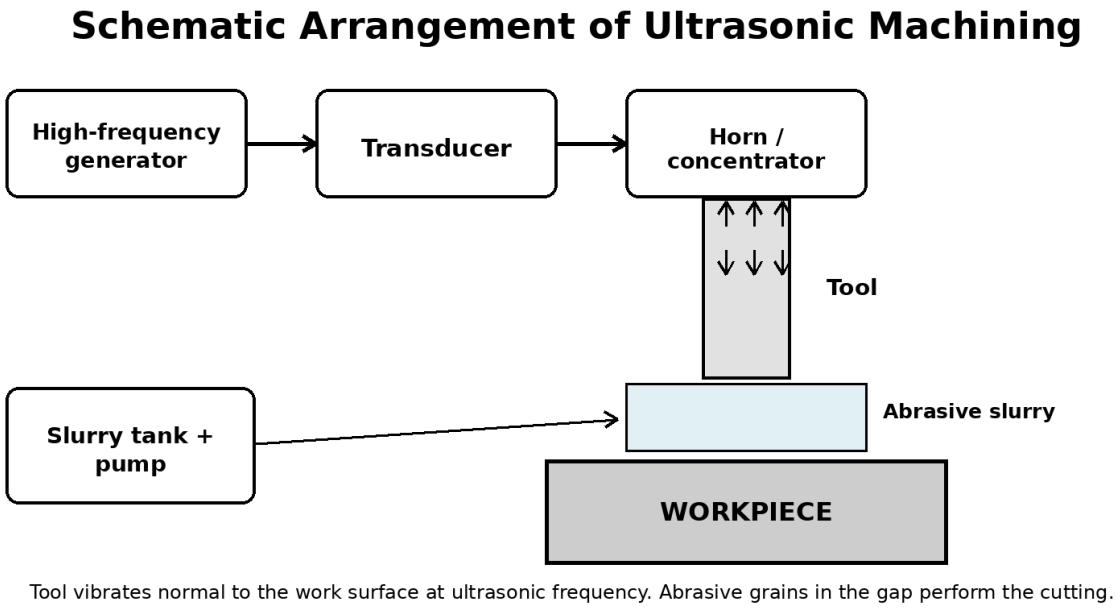


Figure 1. Schematic arrangement of Ultrasonic Machining

2.3 Main Elements of an USM Setup

Element	Function
High-frequency generator	Supplies electrical power at ultrasonic frequency.
Transducer	Converts electrical energy into mechanical vibration. Magnetostrictive and piezoelectric transducers are commonly used.
Horn / concentrator	Transmits vibration and may amplify the displacement amplitude.
Tool	Has the inverse shape of the required cavity; transmits vibration to abrasive grains.

Abrasive slurry system	Supplies and circulates abrasive grains in liquid through the machining gap.
Feed mechanism	Applies controlled static feed force and maintains the tool-work gap.
Work holding system	Rigidly supports the workpiece and resists vibration.

2.4 Machining Process / Working

1. The workpiece is rigidly fixed and the tool is positioned above the location to be machined.

2. A suitable abrasive slurry is continuously supplied into the narrow gap between tool and workpiece.

3. The transducer converts high-frequency electrical energy into mechanical vibration.

4. The horn transmits and amplifies the vibration to the tool.

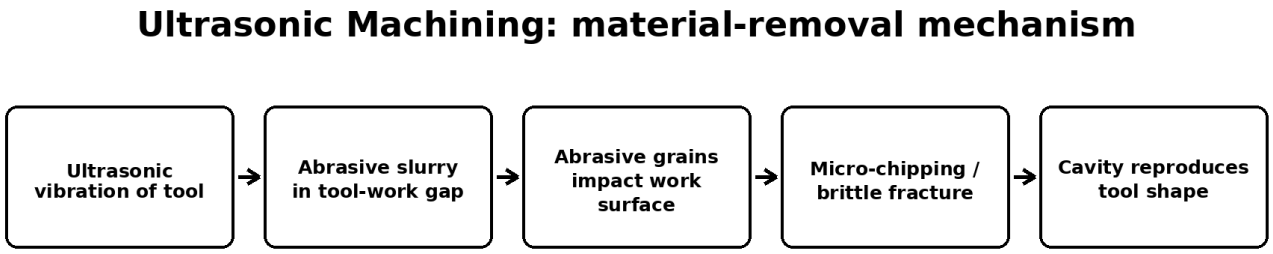
5. The tool oscillates rapidly and drives abrasive grains against the work surface.

6. The grains indent the surface. In brittle material, local tensile stresses initiate cracks and tiny fragments break away.

7. The feed mechanism gradually advances the tool so that the gap remains effective while the cavity becomes deeper.

8. Slurry circulation removes chips and worn abrasive and supplies fresh abrasive grains.

9. Machining continues until the required depth and profile are obtained.



2.5 Material Removal Mechanisms in USM

- Hammering / impact of abrasive particles between the vibrating tool and work surface is the dominant mechanism for brittle materials.
- Micro-chipping and brittle fracture produce the actual fragments removed from glass, ceramics and carbides.
- Abrasive action against the tool also causes tool wear; therefore the tool profile changes slowly during machining.
- Cavitation and local fluid effects can assist debris removal, but they are not normally treated as the principal cutting mechanism.

2.6 Important Process Parameters

Parameter	General influence
Vibration amplitude	Higher amplitude generally increases impact energy and MRR, but excessive amplitude can increase tool wear and surface damage.
Frequency	Affects number of impacts per second and therefore machining rate.
Abrasive size	Coarser grains can increase MRR but usually produce a rougher surface and may reduce accuracy.
Abrasive concentration	There is an effective range; too little gives insufficient cutting action, while too much can restrict grain motion and slurry circulation.
Static feed force	Adequate force maintains abrasive action; excessive force can crush grains or reduce vibration.
Tool material / geometry	Influences wear, dimensional accuracy and ability to transmit vibration.
Slurry flow	Must continuously replenish abrasive and carry away debris.

2.7 Advantages

- Can machine very hard and brittle materials that are difficult to cut conventionally.
- Electrical conductivity of the workpiece is not required.
- Little or no macroscopic cutting force compared with conventional machining; useful for delicate components.
- Complex cavities, small holes and intricate profiles can be produced with a suitably shaped tool.
- No large heat-affected zone because the process is essentially mechanical rather than bulk thermal cutting.
- Good dimensional accuracy and surface finish are possible for suitable applications.
- Can machine non-metals such as glass, quartz and ceramics.

2.8 Disadvantages / Limitations

- Material removal rate is relatively low; it is not an economical bulk-material-removal process.
- Tool wear occurs and can affect dimensional accuracy, especially for deep cavities.
- Deep holes and cavities are difficult because slurry circulation and debris removal become poor.
- Machining of soft and ductile materials is generally less effective than machining brittle materials.
- Equipment requires an ultrasonic generator, transducer and carefully tuned vibration system.
- Abrasive slurry handling, filtration and disposal are required.
- Accuracy can be affected by side cutting and enlargement of the tool profile.

2.9 Applications

- Drilling and profiling glass, quartz, porcelain and advanced ceramics.
- Machining tungsten carbide and other hard tool materials.
- Producing small holes and cavities in brittle electronic and optical materials.
- Engraving, lettering and intricate recesses in hard brittle components.
- Machining dies and inserts where conventional cutting is difficult.
- Producing non-circular holes using shaped tools.

3. Chemical Machining (CHM)

3.1 Fundamental Principle

Chemical Machining removes material by controlled chemical dissolution. Selected areas of the workpiece are exposed to a chemical reagent (etchant), while surfaces that must remain unaffected are protected by a chemically resistant coating called a maskant. The exposed metal reacts with the etchant and is dissolved until the required depth or contour is obtained.

The essential idea is selective corrosion under controlled conditions. CHM does not rely on a cutting edge, spark, or high mechanical force.

3.2 Important Terms

Term	Meaning
Etchant	Chemical solution that attacks and dissolves the exposed work material.
Maskant	Chemical-resistant protective layer applied to areas not to be machined.
Masking	Application of the protective coating.
Scribing / patterning	Removal of maskant from regions where chemical attack is required.
Etch rate	Depth of material removed per unit time under specified conditions.
Undercut	Lateral attack of the etchant beneath the maskant edge; an important source of dimensional error.

3.3 Chemical Machining Process

1. Cleaning and surface preparation: oil, grease, oxide and dirt are removed so that the maskant adheres properly.

2. Masking: a suitable chemically resistant coating is applied over the work surface.

3. Pattern generation: the maskant is removed from the areas where machining is required. This may be done by scribing, cutting, photographic techniques or other patterning methods.

4. Etching: the workpiece is immersed in, sprayed with, or otherwise exposed to the selected etchant. Exposed metal dissolves progressively.

5. Control of depth: machining time, etchant concentration, temperature and agitation are controlled to obtain the required depth.

6. Rinsing and neutralization: the reaction is stopped and residual chemicals are removed.

7. Demasking: remaining maskant is stripped from protected areas.

8. Inspection and finishing: dimensions and surface condition are checked; further cleaning is done if required.

Chemical Machining: basic sequence



3.4 Factors Affecting Chemical Machining

- Work material and its chemical reactivity.
- Nature, concentration and condition of the etchant.
- Temperature: generally affects reaction rate strongly.
- Agitation or circulation of the etchant, which helps remove reaction products.
- Maskant adhesion and chemical resistance.
- Exposure time, which largely controls depth.
- Undercutting at mask edges, which limits dimensional sharpness.

3.5 Advantages

- No cutting forces; thin, delicate and low-stiffness components can be machined without mechanical distortion.
- No tool wear in the conventional sense.
- Large surface areas and many features can be machined simultaneously.
- Complex shallow contours can be produced without expensive shaped cutting tools.
- Burr-free machining is possible because there is no mechanical shearing action.
- Useful for materials that are difficult to machine mechanically but can be attacked chemically.
- Equipment can be simpler than many high-energy machining systems for suitable jobs.

3.6 Disadvantages / Limitations

- Material removal rate and achievable depth are limited; CHM is mainly suited to shallow removal.
- Undercutting reduces dimensional accuracy and makes very sharp corners difficult.
- Surface finish and etch rate depend on material structure and chemical uniformity.
- Chemical handling creates safety, storage, corrosion-control and environmental-disposal requirements.
- Maskant application and pattern preparation can be time-consuming.
- Not every engineering material has a practical, controllable and economical etchant.
- Accurate process control is needed because etch rate changes with concentration, temperature and contamination.

3.7 Applications

- Weight reduction (chemical milling) of aircraft and aerospace panels by removing material from selected regions.
- Production of thin sections, pockets and shallow recesses in sheet and plate.
- Engraving, nameplates, scales and decorative patterns.
- Chemical blanking / photochemical machining of thin metal parts.
- Electronic components and fine patterns where low force and burr-free edges are required.
- Parts having large areas but only small depth of material removal.

4. Electric Discharge Machining (EDM)

4.1 Introduction

Electric Discharge Machining is a thermo-electric non-traditional machining process in which controlled electrical discharges occur across a very small gap between a tool electrode and an electrically conductive workpiece. Each discharge produces intense localized heating. A minute volume of material melts and/or vaporizes, and the dielectric fluid assists in extinguishing the discharge and carrying away debris. Repetition of thousands of controlled discharges produces the required cavity.

4.2 Essential Conditions for EDM

- Both the workpiece and tool electrode must participate in an electrical circuit; the workpiece must therefore be electrically conductive.
- Tool and workpiece are separated by a small controlled gap; normal EDM does not use direct mechanical contact.
- The machining region is filled with a dielectric fluid.
- Electrical energy is supplied as controlled pulses so that individual sparks occur rather than a continuous arc.
- A servo feed system maintains the proper spark gap as material is removed.

4.3 Basic EDM Circuit

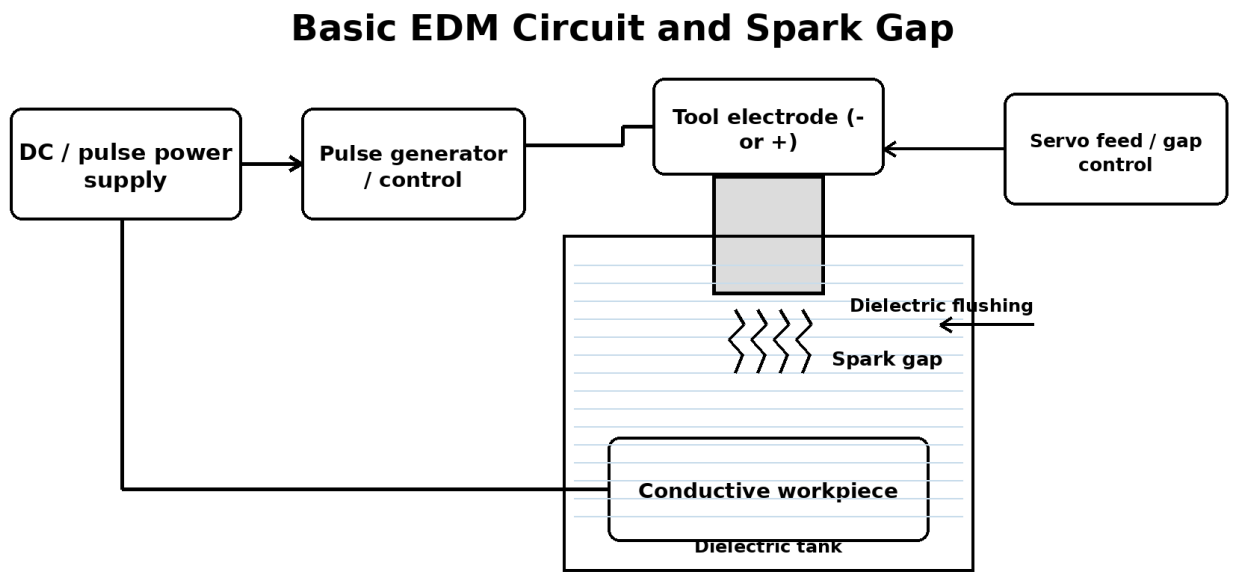


Figure 2. Simplified EDM power/control circuit and spark gap

Part	Function
Power source / pulse generator	Provides controlled voltage and current pulses.
Tool electrode	Forms the negative image of the required cavity and conducts discharge current.
Workpiece	Conductive component from which material is removed.
Dielectric fluid	Insulates until breakdown, cools the gap and flushes

	eroded particles.
Servo feed / gap control	Advances or retracts the electrode to maintain stable sparking.
Tank / flushing arrangement	Contains and circulates dielectric and removes debris.

4.4 Principle of EDM Process

1. Tool and workpiece are positioned very close to each other but do not touch.

2. The dielectric initially behaves as an electrical insulator.

3. When the applied voltage is high enough for the existing gap, the dielectric locally breaks down and an ionized conductive channel forms.

4. A short-duration spark passes through the channel. The temperature at the microscopic discharge spot rises extremely rapidly.

5. A tiny portion of work material melts and may partly vaporize. Some material is also eroded from the electrode.

6. At the end of the pulse, the current is interrupted. The plasma channel collapses and the dielectric regains insulating strength.

7. The rapid cooling and pressure effects help eject molten material, leaving a small crater.

8. The dielectric carries away debris. The servo system readjusts the gap and the next discharge occurs at another microscopic location.

9. Repeated sparks gradually reproduce the tool shape in the workpiece.

4.5 Metal Removal Rate (MRR)

MRR in EDM is the volume (or mass) of work material removed per unit time. Unlike conventional cutting, it is governed primarily by the electrical energy delivered by the discharges and by the efficiency with which molten material is expelled from the gap.

- Peak discharge current: increasing current generally increases discharge energy and MRR, but also increases crater size, roughness and often tool wear.
- Pulse-on time: longer pulse duration can increase energy per spark up to an effective range; excessive duration may promote arcing, poor flushing or an undesirable recast layer.
- Pulse-off time / duty cycle: sufficient off-time is required for deionization and debris removal. Too little off-time causes unstable machining.
- Gap voltage and gap condition: influence initiation and stability of discharges.
- Electrode and work materials: thermal conductivity, melting temperature and other properties affect erosion.
- Dielectric condition and flushing: clean dielectric and effective debris removal permit stable sparking and higher practical MRR.
- Electrode polarity: influences relative erosion of tool and workpiece and is selected according to material and operating conditions.

There is no single universal MRR equation valid for every EDM machine and material combination. In practice, MRR is correlated with pulse energy, average current, duty factor, electrode/work materials, polarity and flushing conditions.

4.6 Dielectric Fluid

The dielectric is an essential working medium in EDM. Its job is not merely cooling; it controls when and where electrical breakdown occurs and strongly influences process stability.

Required function / property	Why it matters
Electrical insulation before breakdown	Prevents continuous current and allows discrete controlled sparks.
Rapid deionization after a pulse	Helps extinguish the discharge so the next pulse can be controlled.
Cooling	Removes heat from tool, workpiece and gap.
Flushing ability	Carries away eroded particles and prevents bridging / short circuiting.
Low viscosity	Improves circulation through narrow spark gaps.
Chemical stability and suitable flash point	Improves safety and service life.
Low contamination / filtration compatibility	Maintains consistent breakdown behavior and stable machining.

Hydrocarbon EDM oils and specially formulated dielectric fluids are common in die-sinking EDM. Deionized water is widely used in wire EDM systems. Fluid selection depends on machine type, work material, required finish, safety and flushing requirements.

4.7 Tool Electrode Materials

- Copper: good electrical/thermal conductivity and good machinability; widely used.
- Graphite: commonly used because it can be machined into complex shapes and has favorable EDM characteristics.
- Copper-tungsten / tungsten-based materials: used for special high-wear or precision applications.
- Electrode material is selected with consideration of tool wear, surface finish, MRR, geometry and cost.

4.8 Advantages

- Can machine very hard conductive materials irrespective of their conventional machinability.
- Can produce intricate cavities, sharp internal features (within spark-gap limitations), small holes and complex dies.
- No large mechanical cutting force; useful for fragile or slender conductive parts.
- No conventional cutting-tool edge is required.
- Good dimensional accuracy and fine surface finish are possible with finishing parameters.
- Can machine hardened parts directly, often reducing the need for post-machining heat treatment operations.
- Burr formation is generally small compared with mechanical cutting.

4.9 Disadvantages / Limitations

- Applicable only to electrically conductive work materials.
- MRR is usually lower than heavy conventional rough machining.
- Tool electrode wear occurs and must be considered in dimensional control.
- Electrical power consumption and equipment cost can be significant.

- A recast / resolidified layer and heat-affected surface may form; microcracks are possible under aggressive conditions.
- Dielectric handling, filtration, fire safety and ventilation may be necessary.
- Accurate electrode manufacture is required for die-sinking EDM.
- Deep narrow cavities can be difficult to flush effectively.

4.10 Applications

- Die and mould cavities, forging dies and extrusion dies.
- Punches and intricate profiles in hardened tool steels.
- Small and deep holes, including holes in hard alloys.
- Removal of broken drills, taps and studs from valuable components.
- Machining carbides and difficult-to-machine conductive alloys.
- Precision features in aerospace, automotive, tool-room and medical components.
- Wire EDM cutting of complex two-dimensional and ruled profiles in conductive materials.

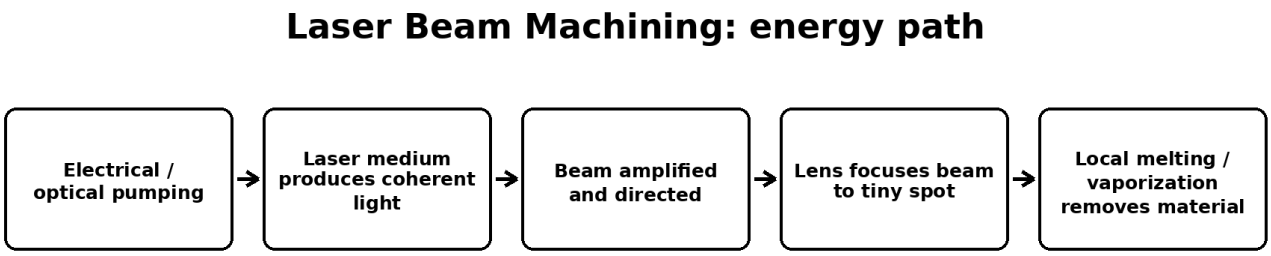
5. Laser Beam Machining (LBM)

5.1 Introduction

Laser Beam Machining is a thermal, non-contact machining process in which a highly concentrated beam of coherent light is focused onto a small area of the workpiece. The very high power density raises the local temperature rapidly, causing melting, vaporization or a combination of both. The molten/vaporized material is expelled from the machining zone, producing a hole, slot or cut.

5.2 Laser Principle in Brief

- Laser means Light Amplification by Stimulated Emission of Radiation.
- An active medium is supplied with energy (pumping) to create a population inversion.
- Stimulated emission produces photons that are coherent and highly directional.
- An optical resonator amplifies the radiation, and an output beam is directed toward the workpiece.
- A lens focuses the beam to a very small spot, producing a high energy density.



5.3 Machining Process / Working

1. The laser source is energized and the active medium produces an intense coherent beam.

2. Mirrors / beam delivery optics direct the beam toward the machining head.

3. A focusing lens concentrates the energy onto a small spot on the work surface.

4. The surface temperature rises rapidly. Depending on power density, pulse duration and material, the material melts and/or vaporizes.

5. Vapour pressure and, in many systems, an assist gas help remove molten material from the interaction zone.

6. The beam or workpiece is moved according to the required path for cutting, drilling, trimming or marking.

7. For pulsed drilling, repeated pulses deepen the hole until breakthrough or the required depth is reached.

5.4 Important Parameters

Parameter	Importance
Laser power / pulse energy	Controls energy delivered and therefore penetration and removal rate.
Pulse duration and repetition rate	Influence peak power, thermal diffusion, hole quality and productivity.
Spot size / focus position	Determines power density and feature size.
Beam quality	Affects ability to focus to a small, consistent spot.
Cutting speed / dwell time	Controls heat input per unit length.
Assist gas	Helps eject molten material and can influence oxidation / cut quality.
Material absorptivity and thermal properties	Strongly affect how efficiently laser energy becomes heat in the workpiece.

5.5 Advantages

- Non-contact process; no mechanical cutting force and no conventional tool wear.
- Can produce very small holes, narrow cuts and fine features.
- High energy density permits machining of many hard, refractory and difficult-to-machine materials.
- Easy to automate and integrate with CNC / computer-controlled motion.
- Can access locations that are difficult for conventional cutting tools when line-of-sight beam delivery is possible.
- High processing speed is possible for thin sections and suitable materials.
- No electrode preparation is required, unlike die-sinking EDM.

5.6 Disadvantages / Limitations

- High initial cost of laser source, optics, controls and safety enclosure.
- Process efficiency can be low because only part of the input energy reaches and removes work material.
- Thermal damage is possible: heat-affected zone, recast, oxidation, microcracks or changes in metallurgical structure.
- Reflective materials can be difficult for some laser wavelengths and operating conditions.
- Accuracy and cut quality deteriorate when thickness becomes large relative to the capability of the laser system.
- Optics require cleanliness, alignment and maintenance.
- Strict eye and skin safety precautions are necessary; fumes and vapours may require extraction.

5.7 Applications

- Micro-drilling of small holes in thin sheets and precision components.
- Cutting and trimming sheet metal, polymers, ceramics and other materials using suitable laser systems.
- Drilling cooling holes and precision apertures in aerospace components.
- Electronic, medical and instrument components requiring small features.
- Marking, engraving and identification.
- Trimming of delicate parts where mechanical forces are undesirable.
- Processing of hard or refractory materials where conventional tools wear rapidly.

6. Comparison of the Four Processes

Feature	USM	CHM	EDM	LBM
Material removal	Abrasive impact / micro-chipping	Chemical dissolution	Spark melting / vaporization	Laser heating, melting / vaporization
Tool contact	No cutting contact	No	No	No
Conductivity needed?	No	No	Yes	No
Best suited to	Hard, brittle materials	Shallow large-area removal	Hard conductive materials / intricate cavities	Fine holes, cuts, trimming, broad material range
Major wear issue	Tool wear	Maskant / chemical control rather than tool wear	Electrode wear	No conventional tool wear; optics need maintenance
Thermal damage	Very small	Negligible thermal effect	Possible recast / HAZ	Possible HAZ / recast
Key limitation	Low MRR	Undercut / chemical handling	Conductive materials only	Equipment cost / thermal effects

7. Important Points

- USM: vibrating tool + abrasive slurry + brittle fracture; excellent for glass, ceramics and carbides.
- CHM: mask unwanted areas + expose required area + chemical etching; controlled corrosion is the removal mechanism.
- EDM: conductive tool and workpiece separated by dielectric; pulsed sparks create microscopic craters.
- EDM MRR increases with discharge energy/current up to practical limits, but roughness, tool wear and thermal damage also increase.
- Dielectric fluid in EDM provides insulation, deionization, cooling and flushing.
- LBM: focused coherent light gives very high power density; removal is predominantly thermal.
- USM and CHM generally avoid significant heat-affected zones; EDM and LBM are thermal processes and can modify the surface layer.

8. Typical Examination Questions

1. Explain the principle and working of Ultrasonic Machining with a neat sketch. State its advantages, limitations and applications.

2. Why is USM particularly suitable for hard and brittle materials? Explain the role of abrasive slurry.

3. Explain the fundamental principle and complete sequence of Chemical Machining. What is undercut?

4. Draw and explain the basic EDM circuit. Explain how an individual spark removes material.

5. Discuss the factors controlling MRR in EDM and state the functions of dielectric fluid.

6. List the advantages, disadvantages and industrial applications of EDM.

7. Explain the principle and working of Laser Beam Machining. Why is it called a non-contact thermal process?

8. Compare USM, CHM, EDM and LBM with respect to energy source, work material, tool wear, thermal damage and applications.

9. Reference

Primary requested reference framework: S. K. Hajra Choudhury, S. K. Bose and A. K. Hajra Choudhury, Elements of Workshop Technology, Vol. II, Media Promoters & Publishers. The topics and organization also align with engineering syllabi that list this text for advanced/non-conventional machining.